

leaderboard_snapshots	bigserial
snapshot_id ♂	
leaderboard_type	varchar(50) NN
class_level	integer
school_name	varchar(255)
period	varchar(28)
snapshot_date	date NN
rankings	jsonb NN
total_users	integer
last_updated_at	timestamp

boards	serial
board_id ♂	
board_name ♂	varchar(180) NN
board_code ♂	varchar(20) NN
description	text
is_active	boolean
display_order	integer
metadata	jsonb
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

exam_categories	serial
category_id ♂	
category_name ♂	varchar(180) NN
category_code ♂	varchar(50) NN
description	text
icon_url	varchar(580)
is_active	boolean
display_order	integer
metadata	jsonb
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

difficulty_levels	serial
difficulty_id ♂	
difficulty_name ♂	varchar(180) NN
difficulty_code ♂	varchar(20) NN
description	text
points_multiplier	decimal(12,2)
display_order	integer
color_hex	varchar(17)
is_active	boolean
created_at	timestamp
updated_at	timestamp

question_types	serial
type_id ♂	
type_name ♂	varchar(180) NN
type_code ♂	varchar(20) NN
description	text
validation_rules	jsonb
scoring_rules	jsonb
is_active	boolean
display_order	integer
created_at	timestamp
updated_at	timestamp

users	bigserial
user_id ♂	
email ♂	varchar(255) NN
password_hash	varchar(255) NN
role	varchar(28) NN
status	varchar(28) NN
email_verified	boolean
email_verification_token	varchar(255)
password_reset_token	varchar(255)
password_reset_expires_at	timestamp
last_login_at	timestamp
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

subjects	serial
subject_id ♂	
subject_name	varchar(180) NN
subject_code ♂	varchar(50) NN
description	text
icon_url	varchar(580)
color_hex	varchar(17)
is_active	boolean
display_order	integer
metadata	jsonb
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

tags	serial
tag_id ♂	
tag_name ♂	varchar(180) NN
tag_category	varchar(50)
description	text
is_active	boolean
created_at	timestamp

badges	serial
badge_id ♂	
badge_name ♂	varchar(180) NN
badge_code ♂	varchar(50) NN
description	text
icon_url	varchar(580)
criteria	jsonb NN
category	varchar(50)
class_range	jsonb
rarity	varchar(28)
display_order	integer
is_active	boolean
created_at	timestamp
updated_at	timestamp

user_profiles	bigserial
profile_id ♂	
user_id ♂ ♂	bigint NN
full_name	varchar(255) NN
school_name	varchar(255)
class_level	integer NN
board_id ♂	integer
parent_name	varchar(255)
parent_email	varchar(280)
parent_phone	varchar(28)
date_of_birth	date
avatar_url	varchar(580)
bio	text
preferences	jsonb
metadata	jsonb
created_at	timestamp
updated_at	timestamp

user_sessions	bigserial
session_id ♂	
user_id ♂	bigint NN
refresh_token ♂	varchar(580) NN
device_info	jsonb
ip_address	inet
expires_at	timestamp NN
created_at	timestamp

topics	serial
topic_id ♂	
subject_id ♂	integer NN
parent_topic_id ♂	integer
topic_name	varchar(280) NN
topic_code	varchar(50)
description	text
class_level	integer
display_order	integer
is_active	boolean
metadata	jsonb
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

tests	bigserial
test_id ♂	
test_name	varchar(255)
test_type	varchar(28) NN
test_config	jsonb NN
subject_id ♂	integer
exam_category_id ♂	integer
class_level	integer
access_level	varchar(28)
points_cost	boolean
is_template	boolean
is_active	boolean
created_by ♂	bigint
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

points_transactions	bigserial
transaction_id ♂	
user_id ♂	bigint NN
points	integer NN
transaction_type	varchar(50) NN
reference_type	varchar(50)
reference_id	bigint
balance_after	integer NN
description	text
metadata	jsonb
created_at	timestamp

subscription_plans	serial
plan_id ♂	
plan_name	varchar(180) NN
plan_code ♂	varchar(50) NN
description	text
price_per_month	decimal(14,2) NN
price_per_quarter	decimal(14,2)
price_per_year	decimal(14,2)
class_level	integer
subject_id ♂	integer
features	jsonb
is_active	boolean
display_order	integer
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

configuration_audit_log	bigserial
log_id ♂	
table_name	varchar(180) NN
record_id	integer
action	varchar(28) NN
old_values	jsonb
new_values	jsonb
changed_fields	jsonb
changed_by ♂	bigint NN
changed_at	timestamp
ip_address	inet
user_agent	text

user_badges	bigserial
user_badge_id ♂	
user_id ♂	bigint NN
badge_id ♂	integer NN
earned_at	timestamp
metadata	jsonb

questions	bigserial
question_id ♂	
question_text	text NN
question_html	text
question_image_url	varchar(580)
type_id ♂	integer NN
difficulty_id ♂	integer NN
subject_id ♂	integer NN
topic_id ♂	integer
class_level	integer NN
exam_category_id ♂	integer NN
correct_answer	text
correct_answer_data	jsonb
time_limit_seconds	integer
marks	decimal(5,2)
negative_marks	decimal(5,2)
times_attempted	integer
times_correct	integer
success_rate	decimal(5,2)
avg_time_taken_seconds	integer
tags	jsonb
metadata	jsonb
status	varchar(28)
is_active	boolean
created_by ♂	bigint
updated_by ♂	bigint
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

test_attempts	bigserial
attempt_id ♂	
test_id ♂	bigint NN
user_id ♂	bigint NN
status	varchar(28)
started_at	timestamp
submitted_at	timestamp
time_taken_seconds	integer
total_questions	integer NN
attempted_questions	integer
correct_answers	integer
wrong_answers	integer
unattempted	integer
score	decimal(6,2)
max_score	decimal(6,2) NN
percentage	decimal(5,2)
responses_data	jsonb
metadata	jsonb
created_at	timestamp
updated_at	timestamp

payments	bigserial
payment_id ♂	
user_id ♂	bigint NN
amount	decimal(10,2) NN
currency	varchar(3)
payment_method	varchar(50)
gateway_transaction_id ♂	varchar(255)
gateway_name	varchar(50)
gateway_response	jsonb
status	varchar(28)
plan_id ♂	integer
subscription_id ♂	bigint
metadata	jsonb
created_at	timestamp
updated_at	timestamp
completed_at	timestamp

question_options	bigserial
option_id ♂	
question_id ♂	bigint NN
option_text	text NN
option_html	text
option_image_url	varchar(580)
option_order	integer NN
is_correct	boolean
metadata	jsonb
created_at	timestamp
updated_at	timestamp

question_explanations	bigserial
explanation_id ♂	
question_id ♂	bigint NN
explanations_type	varchar(28)
explanation_text	text NN
explanation_html	text
explanation_images	jsonb
explanation_video_url	varchar(580)
step_by_step	jsonb
display_order	integer
created_at	timestamp
updated_at	timestamp

question_tags	bigint
question_id ♂ ♂	
tag_id ♂ ♂	integer
created_at	timestamp

test_questions	bigserial
test_question_id ♂	
test_id ♂	bigint NN
question_id ♂	bigint NN
question_order	integer NN
marks	decimal(5,2)
negative_marks	decimal(5,2)
created_at	timestamp

user_subscriptions	bigserial
subscription_id ♂	
user_id ♂	bigint NN
plan_id ♂	integer NN
start_date	date NN
end_date	date NN
status	varchar(28)
auto_renewal	boolean
payment_id ♂	bigint
metadata	jsonb
created_at	timestamp
updated_at	timestamp
cancelled_at	timestamp

question_reports	bigserial
report_id ♂	
question_id ♂	bigint NN
reported_by ♂	bigint NN
reported_type	varchar(50) NN
description	text NN
status	varchar(28)
resolved_by ♂	bigint
resolution_notes	text
resolved_at	timestamp
metadata	jsonb
created_at	timestamp
updated_at	timestamp

test_responses	bigserial
response_id ♂	
attempt_id ♂	bigint NN
question_id ♂	bigint NN
selected_option_id ♂	bigint
selected_option_ids	jsonb
test_answer	text
is_correct	boolean
marks_awarded	decimal(5,2)
marks_possible	decimal(5,2) NN
time_taken_seconds	integer
marked_for_review	boolean
response_order	integer
metadata	jsonb
created_at	timestamp
updated_at	timestamp

user_subject_access	bigserial
access_id ♂	
user_id ♂	bigint NN
subject_id ♂	integer NN
access_type	varchar(20) NN
free_questions_used	integer
free_questions_limit	integer
granted_at	timestamp
expires_at	timestamp
subscription_id ♂	bigint
metadata	jsonb

leaderboard_snapshots	
snapshot_id	bigserial
leaderboard_type	varchar(50) NN
class_level	integer
school_name	varchar(255)
period	varchar(20)
snapshot_date	date NN
rankings	jsonb NN
total_users	integer
last_updated_at	timestamp

boards	
board_id 	serial 
board_name 	varchar(100) NN
board_code	varchar(20) NN
description	text
is_active	boolean
display_order	integer
metadata	jsonb
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

exam_categories	
category_id 	serial
category_name 	varchar(100) NN
category_code 	varchar(50) NN
description	text
icon_url	varchar(500)
is_active	boolean
display_order	integer
metadata	jsonb
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

difficulty_levels	
difficulty_id $\emptyset$	serial
difficulty_name $\emptyset$	varchar(50) NN
difficulty_code $\emptyset$	varchar(20) NN
description	text
points_multiplier	decimal(1,2)
display_order	integer
color_hex	varchar(7)
is_active	boolean
created_at	timestamp
updated_at	timestamp

question_types	
type_id $\varnothing$	serial 
type_name $\emptyset$	varchar(50) NN
type_code $\emptyset$	varchar(20) NN
description	text
validation_rules	jsonb
scoring_rules	jsonb
is_active	boolean
display_order	integer
created_at	timestamp
updated_at	timestamp

users	
user_id $\varnothing$	bigserial
email $\emptyset$	varchar(255) NN
password_hash	varchar(255) NN
role	varchar(20) NN
status	varchar(20) NN
email_verified	boolean
email_verification_token	varchar(255)
password_reset_token	varchar(255)
password_reset_expires_at	timestamp
last_login_at	timestamp
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

subjects	
subject_id	serial
subject_name	varchar(100) NN
subject_code	varchar(50) NN
description	text
icon_url	varchar(500)
color_hex	varchar(7)
is_active	boolean
display_order	integer
metadata	jsonb
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

tags	
tag_id	serial
tag_name	varchar(50) NN
tag_category	varchar(50)
description	text
is_active	boolean
created_at	timestamp

badges		1
badge_id	serial	
badge_name	varchar(100)	NN
badge_code	varchar(50)	NN
description	text	
icon_url	varchar(500)	
criteria	jsonb	NN
category	varchar(50)	
class_range	jsonb	
rarity	varchar(20)	
display_order	integer	
is_active	boolean	
created_at	timestamp	
updated_at	timestamp	

user_profiles	
profile_id	bigserial
user_id	bigint NN
full_name	varchar(255) NN
school_name	varchar(255)
class_id	integer NN
board_id	integer
parent_name	varchar(255)
parent_email	varchar(255)
parent_phone	varchar(20)
date_of_birth	date
avatar_url	varchar(500)
bio	text
preferences	jsonb
metadata	jsonb
created_at	timestamp
updated_at	timestamp

user_sessions	
session_id	bigserial
user_id	bigint NN
refresh_token	varchar(500) NN
device_info	jsonb
ip_address	inet
expires_at	timestamp NN
created_at	timestamp

topics	
topic_id	serial
subject_id	integer NN
parent_topic_id	integer
topic_name	varchar(200) NN
topic_code	varchar(50)
description	text
class_level	integer
display_order	integer
is_active	boolean
metadata	jsonb
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

tests	
test_id $\varnothing$	bigserial
test_name	varchar(255)
test_type	varchar(20) NN
test_config	jsonb NN
subject_id $\varnothing$	integer
exam_category_id $\varnothing$	integer
class_level	integer
access_level	varchar(20)
points_cost	integer
is_template	boolean
is_active	boolean
created_by $\varnothing$	bigint
created_at	timestamp
updated_at	timestamp
deleted_at	timestamp

points_transactions	
transaction_id $\emptyset$	bigserial
user_id $\emptyset$	bigint NN
points	integer NN
transaction_type	varchar(50) NN
reference_type	varchar(50)
reference_id	bigint
balance_after	integer NN
description	text
metadata	jsonb
created_at	timestamp

subscription_plans		0
plan_id 	serial	1
plan_name	varchar(100) NN	
plan_code 	varchar(50) NN	
description	text	
price_per_month	decimal(10,2) NN	
price_per_quarter	decimal(10,2)	
price_per_year	decimal(10,2)	
class_level	integer	
subject_id 	integer	
features	jsonb	
is_active	boolean	
display_order	integer	
created_at	timestamp	
updated_at	timestamp	
deleted_at	timestamp	

configuration_audit_log	
log_id 	bigserial
table_name	varchar(100) 
record_id	integer 
action	varchar(20) 
old_values	jsonb
new_values	jsonb
changed_fields	jsonb
changed_by	bigint
changed_at	timestamp
ip_address	inet
user_agent	text

user_badges	
user_badge_id 	bigserial
user_id 	bigint NN
badge_id	integer NN
earned_at	timestamp
metadata	jsonb

questions		1
question_id $\varnothing$	bigserial	
question_text	text	NN
question_html	text	
question_image_url	varchar(500)	
type_id $\varnothing$	integer	NN
difficulty_id $\varnothing$	integer	NN
subject_id $\varnothing$	integer	NN
topic_id $\varnothing$	integer	
class_level	integer	NN
exam_category_id $\varnothing$	integer	NN
correct_answer	text	
correct_answer_data	jsonb	
time_limit_seconds	integer	
marks	decimal(5,2)	
negative_marks	decimal(5,2)	
times_attempted	integer	
times_correct	integer	
success_rate	decimal(5,2)	
avg_time_taken_seconds	integer	
tags	jsonb	
metadata	jsonb	
status	varchar(20)	
is_active	boolean	
created_by $\varnothing$	bigint	
updated_by $\varnothing$	bigint	
created_at	timestamp	
updated_at	timestamp	
deleted_at	timestamp	

test_attempts	
attempt_id $\varnothing$	bigserial
test_id $\varnothing$	bigint NN
user_id $\varnothing$	bigint NN
status	varchar(20)
started_at	timestamp
submitted_at	timestamp
time_taken_seconds	integer
total_questions	integer NN
attempted_questions	integer
correct_answers	integer
wrong_answers	integer
unattempted	integer
score	decimal(6,2)
max_score	decimal(6,2) NN
percentage	decimal(5,2)
questions_data	jsonb
responses_summary	jsonb
metadata	jsonb
created_at	timestamp
updated_at	timestamp

payments	
payment_id 	bigserial
user_id 	bigint NN
amount	decimal(10,2) NN
currency	varchar(3)
payment_method	varchar(50)
gateway_transaction_id 	varchar(255)
gateway_name	varchar(50)
gateway_response	json
status	varchar(20)
plan_id 	integer
subscription_id 	bigint
metadata	json
created_at	timestamp
updated_at	timestamp
completed_at	timestamp

question_options	
option_id 	bigserial 
question_id 	bigint NN 
option_text	text NN 
option_html	text
option_image_url	varchar(500)
option_order	integer NN
is_correct	boolean
metadata	jsonb
created_at	timestamp
updated_at	timestamp

question_explanations	
explanation_id 	bigserial
question_id	bigint NN
explanation_type	varchar(20)
explanation_text	text NN
explanation_html	text
explanation_images	jsonb
explanation_video_url	varchar(500)
step_by_step	jsonb
display_order	integer
created_at	timestamp
updated_at	timestamp

question_tags	
question_id	bigint
tag_id	integer
created_at	timestamp

test_questions	
test_question_id	bigserial
test_id	bigint NN
question_id	bigint NN
question_order	integer NN
marks	decimal(5,2)
negative_marks	decimal(5,2)
created_at	timestamp

user_subscriptions	
subscription_id $\varnothing$	bigserial 0..1
user_id $\varnothing$	bigint NN
plan_id $\varnothing$	integer NN
start_date	date NN
end_date	date NN
status	varchar(20)
auto_renewal	boolean
payment_id $\varnothing$	bigint
metadata	jsonb
created_at	timestamp
updated_at	timestamp
cancelled_at	timestamp

question_reports	
report_id 	bigserial
question_id 	bigint NN
reported_by 	bigint NN
report_type	varchar(50) NN
description	text NN
status	varchar(20)
resolved_by 	bigint
resolution_notes	text
resolved_at	timestamp
metadata	jsonb
created_at	timestamp
updated_at	timestamp

test_responses	
response_id 	bigserial
attempt_id 	bigint NN
question_id 	bigint NN
selected_option_id 	bigint
selected_option_ids	jsonb
test_answer	text
is_correct	boolean
marks_awarded	decimal(5,2)
marks_possible	decimal(5,2) NN
time_taken_seconds	integer
marked_for_review	boolean
response_order	integer
metadata	jsonb
created_at	timestamp
updated_at	timestamp

user_subject_access	
access_id 	bigserial
user_id 	bigint NN
subject_id 	integer NN
access_type	varchar(20) NN
free_questions_used	integer
free_questions_limit	integer
granted_at	timestamp
expires_at	timestamp
subscription_id	bigint
metadata	jsonb